

P-100 Series Centrifugal Pump Manual

Operations and Maintenance Manual | Document OEM-P100-OM-06 | Revision 6

1. Equipment scope

This manual applies to Riverbend equipment tags P-101, P-102, and P-103. P-101 is the duty cooling-water pump serving Utilities Cooling Loop A. The drive-end and non-drive-end bearing assemblies use lithium-complex grease under normal service.

2. Normal operating envelope

Parameter	Normal	Alert	Required response
Overall vibration RMS	0-4.5 mm/s	Above 7.1 mm/s	Reduce load and inspect bearing condition
Bearing temperature	55-75 C	Above 85 C	Inspect lubrication and bearing immediately
Discharge pressure	3.6-4.2 bar	Below 3.2 bar	Check suction and impeller condition

If vibration exceeds 7.1 mm/s RMS while bearing temperature is also trending above 85 C, treat the condition as a compound bearing-risk event. Inspect lubrication condition, alignment, and the drive-end bearing before continued operation.

3. Lubrication requirements

Lubricate the drive-end bearing every 30 calendar days or 500 operating hours, whichever occurs first. Do not postpone lubrication beyond seven days. Record work-order ID, grease quantity, technician, and post-maintenance vibration reading.

4. Abnormal-condition troubleshooting

Observed condition	Probable cause	Action
Rising vibration and metallic noise	Bearing wear or lubrication starvation	Stop and inspect drive-end bearing
High temperature without vibration	Excess grease or cooling restriction	Verify grease quantity and airflow
Vibration plus temperature rise	Progressive bearing degradation	Reduce load; inspect within 24 hours

5. Return-to-service criteria

Following bearing or lubrication work, operate at 50 percent load for 20 minutes. Return to full duty only when vibration remains below 4.5 mm/s RMS and bearing temperature stabilises below 75 C. Maintenance clearance must be recorded in the CMMS.